Logic & Circuit Design

Complete Exam Preparation Guide

Table of Contents

1. Propositional Logic
2. Logical Operators
3. Logical Equivalence
4. Predicate Logic
5. Quantifiers
6. Nested Quantifiers
7. Satisfiability
8. Rules of Inference
9. Normal Forms (CNF / DNF)
10. Circuit Design
11. Circuit Minimisation — K-Map
12. **20 Exam Practice Questions**

EXAM READY — MCQ + Written

Truth tables · Equivalences · Quantifiers · Inference rules · K-Maps · Circuit diagrams

Propositional · Predicate · Quantifiers · SAT · Inference · CNF/DNF · Circuits · K-Map

§1 Propositional Logic

▶ Definition: Proposition

A **proposition** (statement) is a declarative sentence that is either **true (T)** or **false (F)**, never both.

Propositions: "$2 + 2 = 4$" ✓ "It is raining" ✓
Not propositions: "What time is it?" × "$x + 1 = 2$" × (open sentence)

▶ Definition: Atomic vs Compound Propositions

- **Atomic:** Cannot be broken down — denoted $p, q, r, \ldots$
- **Compound:** Built from atomic propositions using **logical connectives**

▶ Definition: Tautology, Contradiction, Contingency

- **Tautology:** Always true (e.g. $p \vee \neg p$)
- **Contradiction:** Always false (e.g. $p \wedge \neg p$)
- **Contingency:** Sometimes true, sometimes false

§2 Logical Operators

0.1 The Six Core Connectives

Name	Symbol	Read As	True when...
Negation	$\neg p$	"not p"	p is false
Conjunction	$p \wedge q$	"p and q"	both true
Disjunction	$p \vee q$	"p or q"	at least one true
Exclusive OR	$p \oplus q$	"p xor q"	exactly one true
Implication	$p \to q$	"if p then q"	not (T→F)
Biconditional	$p \leftrightarrow q$	"p iff q"	same truth value

0.2 Complete Truth Table

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	**F**	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

★ Key Exam Point: Implication is the Tricky One

$p \to q$ is **FALSE only when p is TRUE and q is FALSE**.
"If it rains, I carry an umbrella" is only broken if it rains AND I have no umbrella.

Also remember: $p \to q \equiv \neg p \vee q$

0.3 Operator Precedence (high to low)

5. $\neg$ (NOT)
4. $\wedge$ (AND)
3. $\vee$ (OR)
2. $\to$ (IMPLIES)
1. $\leftrightarrow$ (IFF)

precedence

Example: Parsing with Precedence

$p \wedge q \to \neg r \vee s$ is parsed as $((p \wedge q) \to ((\neg r) \vee s))$

§3 Logical Equivalence

▶ Definition: Logical Equivalence

$p \equiv q$ (or $p \Leftrightarrow q$) if p and q have **identical truth tables** — i.e. the biconditional $p \leftrightarrow q$ is a tautology.

Theorem / Law: Key Equivalence Laws

Law	Form 1	Form 2
Identity	$p \wedge \mathbf{T} \equiv p$	$p \vee \mathbf{F} \equiv p$
Domination	$p \vee \mathbf{T} \equiv \mathbf{T}$	$p \wedge \mathbf{F} \equiv \mathbf{F}$
Idempotent	$p \wedge p \equiv p$	$p \vee p \equiv p$
Double Neg.	$\neg(\neg p) \equiv p$	
Commutative	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	(same for $\vee$)
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	(and dual)
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Implication	$p \to q \equiv \neg p \vee q$	
Contrapositive	$p \to q \equiv \neg q \to \neg p$	

Example: De Morgan's Law — Verify

Show $\neg(p \wedge q) \equiv \neg p \vee \neg q$:

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$	Equal?
T	T	T	F	F	✓
T	F	F	T	T	✓
F	T	F	T	T	✓
F	F	F	T	T	✓

★ Key Exam Point: De Morgan's — Don't Forget

$\neg(p \wedge q) \neq \neg p \wedge \neg q$ (wrong!)
You must **flip both operands AND flip the connective**: AND $\leftrightarrow$ OR.

§4 Predicate Logic

▶ Definition: Predicate (Propositional Function)

A **predicate** $P(x)$ is a statement containing a variable x.
It becomes a proposition when x is assigned a value or bound by a quantifier.

Example: $P(x) =$ "$x > 3$" $P(5)$ is TRUE, $P(2)$ is FALSE.

▶ Definition: Domain of Discourse

The **domain** (universe of discourse) is the set of allowed values for variables.
The truth of $P(x)$ can depend on the domain!

Example: Predicate vs Proposition

- $P(x) =$ "$x^2 > 0$" — predicate (depends on x)
- $P(3) =$ "$9 > 0$" — proposition (TRUE)
- $\forall x\, P(x)$ — proposition (depends on domain)

§5 Quantifiers

▶ Definition: Universal Quantifier $\forall$

$\forall x\, P(x)$ means "**For all** x in the domain, $P(x)$ is true."
It is **FALSE** if even one counterexample exists.

▶ Definition: Existential Quantifier $\exists$

$\exists x\, P(x)$ means "**There exists** some x in the domain for which $P(x)$ is true."
It is **TRUE** if at least one witness exists.

▶ Definition: Unique Existence $\exists!$

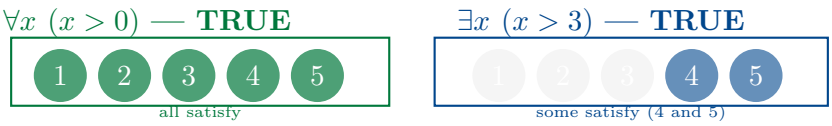
$\exists!\, x\, P(x)$ means "There exists **exactly one** x with $P(x)$."

Theorem / Law: Negation of Quantifiers (De Morgan for Quantifiers)

$$\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$$
$$\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$$

"Not all" $\equiv$ *"some don't"* *"None"* $\equiv$ *"all don't"*

Quantifier Truth Visual (domain $= \{1, 2, 3, 4, 5\}$)



★ Key Exam Point: Quantifier Scope

$\forall x\, (P(x) \to Q(x))$ is **not** the same as $(\forall x\, P(x)) \to (\forall x\, Q(x))$.
The quantifier binds the entire formula in its scope!

§6 Nested Quantifiers

▶ Definition: Nested Quantifiers

Multiple quantifiers applied to a predicate of several variables:
$\forall x\, \exists y\, P(x, y)$ "For every x, there is some y such that $P(x, y)$."

Order matters! $\forall x\, \exists y \not\equiv \exists y\, \forall x$ in general.

Order of Quantifiers — Critical Table

Statement	Meaning	Example (integers)
$\forall x\, \forall y\, P(x, y)$	For all x, for all y	$\forall x\, \forall y\, (x + y = y + x)$ — TRUE (commutativity)
$\exists x\, \exists y\, P(x, y)$	There exist x and y	$\exists x\, \exists y\, (x + y = 5)$ — TRUE
$\forall x\, \exists y\, P(x, y)$	For every x, some y works	$\forall x\, \exists y\, (x + y = 0)$ — TRUE ($y = -x$)
$\exists x\, \forall y\, P(x, y)$	One x works for all y	$\exists x\, \forall y\, (x \cdot y = 0)$ — TRUE ($x = 0$)

Example: Translating English ↔ Logic

- "Every student has a teacher" $\equiv \forall s\, \exists t\, \text{Teaches}(t, s)$
- "Some teacher teaches all students" $\equiv \exists t\, \forall s\, \text{Teaches}(t, s)$
- "No one is their own teacher" $\equiv \forall x\, \neg\text{Teaches}(x, x)$

Theorem / Law: Negating Nested Quantifiers

Push $\neg$ inward, flipping each quantifier:

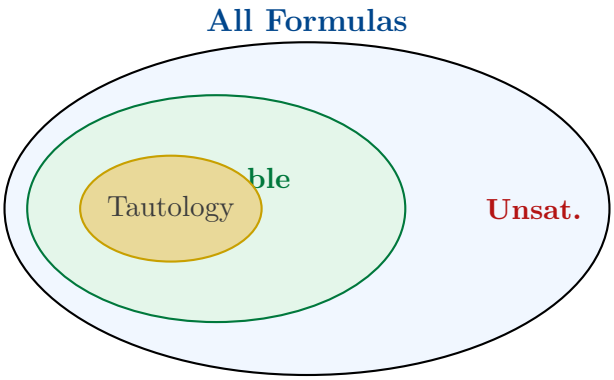
$$\neg \forall x \, \exists y \, P(x, y) \;\equiv\; \exists x \, \forall y \, \neg P(x, y)$$

§7 Satisfiability

▶ Definition: Satisfiability

A propositional formula ϕ is **satisfiable** if there exists at least one truth assignment to its variables that makes ϕ **TRUE**.

- **Satisfiable** — at least one row in truth table is T
- **Unsatisfiable (contradiction)** — all rows are F
- **Valid (tautology)** — all rows are T



Example: SAT Checking

Is $\phi = (p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee \neg r)$ satisfiable?
Try $p = \text{T}, q = \text{F}, r = \text{T}$: $(T) \wedge (T) \wedge (T) = \text{TRUE}$. **Satisfiable!**

Note: SAT is NP-Complete

The Boolean Satisfiability Problem (SAT) is the first proven NP-complete problem (Cook, 1971). No known polynomial-time algorithm exists.

§8 Rules of Inference

▶ Definition: Argument & Valid Argument

An **argument** is a sequence of propositions (premises) leading to a conclusion.
It is **valid** if whenever all premises are true, the conclusion must be true.

$$\frac{P_1 \quad P_2 \quad \cdots \quad P_n}{\therefore\; C}$$

Rules of Inference — Master Table

Rule	Form	Example
Modus Ponens	$p,\ p \to q\ \therefore\ q$	"It rains; rain → wet; ∴ wet"
Modus Tollens	$\neg q,\ p \to q\ \therefore\ \neg p$	"Not wet; rain → wet; ∴ no rain"
Hypothetical Syll.	$p \to q,\ q \to r\ \therefore\ p \to r$	Chain of implications
Disjunctive Syll.	$p \vee q,\ \neg p\ \therefore\ q$	"A or B; not A; ∴ B"
Addition	$p\ \therefore\ p \vee q$	Weaken a fact
Simplification	$p \wedge q\ \therefore\ p$	Drop one conjunct
Conjunction	$p,\ q\ \therefore\ p \wedge q$	Combine two facts
Resolution	$p \vee q,\ \neg p \vee r\ \therefore\ q \vee r$	SAT/theorem provers

Example: Formal Argument with Rules

Premises: "If it snows, roads are icy." "Roads are not icy." Conclusion: "It did not snow."

Let s = snows, i = icy.
Premises: $s \to i,\quad \neg i$
By **Modus Tollens**: $\neg i,\ s \to i\ \therefore\ \neg s$. ✓

★ Key Exam Point: Fallacies to Avoid

- **Affirming the consequent:** $q,\ p \to q\ \therefore\ p$ **INVALID**
- **Denying the antecedent:** $\neg p,\ p \to q\ \therefore\ \neg q$ **INVALID**

§9 Normal Forms (CNF & DNF)

▶ Definition: Literal

A **literal** is a variable (p) or its negation ($\neg p$).

▶ Definition: Conjunctive Normal Form (CNF)

A formula is in **CNF** if it is a **conjunction of clauses**, where each clause is a **disjunction of literals**:

$$(l_{11} \vee l_{12} \vee \cdots) \wedge (l_{21} \vee l_{22} \vee \cdots) \wedge \cdots$$

"AND of ORs" — used in SAT solvers.

▶ Definition: Disjunctive Normal Form (DNF)

A formula is in **DNF** if it is a **disjunction of terms**, where each term is a **conjunction of literals**:

$$(l_{11} \wedge l_{12} \wedge \cdots) \vee (l_{21} \wedge l_{22} \wedge \cdots) \vee \cdots$$

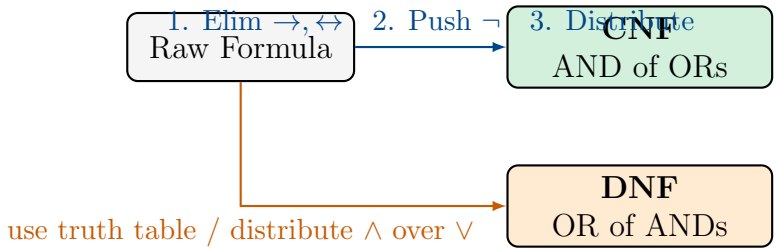
"OR of ANDs" — corresponds directly to rows of a truth table that are TRUE.

Example: Converting to DNF and CNF

Formula: $(p \to q) \wedge r$

Step 1: Eliminate $\to$: $(\neg p \vee q) \wedge r$
This is already CNF: $(\neg p \vee q) \wedge (r)$

For DNF: Distribute: $(\neg p \wedge r) \vee (q \wedge r)$ ✓



★ Key Exam Point: CNF vs DNF — Quick Memory Aid

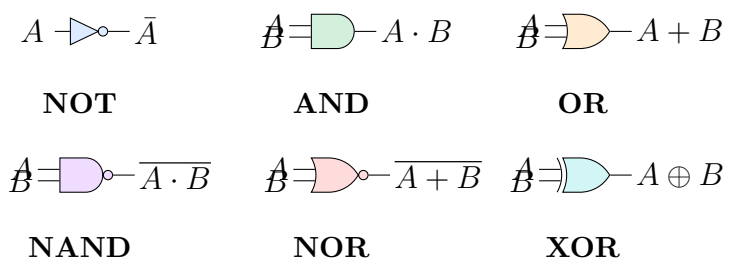
CNF = **C**lauses (OR-groups) ANDed together "**C**onjunction of **C**lauses"
DNF = **D**isjunction of terms (AND-groups) "**D**isjunction of **D**ata-rows"

§10 Circuit Design

▶ Definition: Logic Gate

A **logic gate** implements a boolean function in hardware. Each gate corresponds to a logical connective.

0.4 Gate Symbols and Truth Behaviour

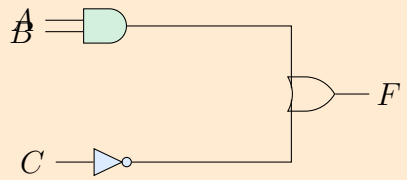


Note: NAND & NOR are Universal Gates

Any boolean function can be built using **only NAND** gates, or **only NOR** gates. This is why they are called **universal** — important for chip manufacturing.

0.5 Building a Circuit from a Boolean Expression

Example: Circuit for $F = (A \wedge B) \vee \neg C$

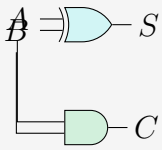


0.6 Half Adder & Full Adder

Half Adder (adds 2 bits)

Inputs: A, B
Outputs: Sum $S = A \oplus B$, Carry $C = A \wedge B$

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



§11 Circuit Minimisation — Karnaugh Map (K-Map)

▶ Definition: K-Map

A **Karnaugh Map** is a visual method to minimise boolean expressions by grouping adjacent minterms. It reduces gates needed in a circuit.

0.7 K-Map Layout Rules

Grouping Rules

1. Groups must be **powers of 2**: 1, 2, 4, 8, ...
2. Groups must be **rectangular** (including wrap-around edges)
3. Use the **largest possible groups**
4. Every "1" cell must be covered by at least one group
5. **Overlapping is allowed**
6. Fewer, larger groups $\Rightarrow$ simpler expression

0.8 2-Variable K-Map

$A \setminus B$	$B = 0$	$B = 1$
$A = 0$	m_0	m_1
$A = 1$	m_2	m_3

0.9 3-Variable K-Map Example

Example: Minimise $F(A, B, C) = \sum m(0, 1, 2, 3, 4, 6)$

	$BC = 00$	$BC = 01$	$BC = 11$	$BC = 10$
$A = 0$	1	1	1	1
$A = 1$	1	0	0	1

Blue group ($A = 0$, all B, C): contributes $\bar{A}$
Green group ($B = 0$, both A): contributes $\bar{B}\bar{C}$
Orange group ($C = 0$, both A, wraps): contributes $\bar{C}$

Minimised: $F = \bar{A} \vee \bar{C}$ (verify covers all 1s)

0.10 4-Variable K-Map Layout

	$CD = 00$	$CD = 01$	$CD = 11$	$CD = 10$
$AB = 00$	0	1	3	2
$AB = 01$	4	5	7	6
$AB = 11$	12	13	15	14
$AB = 10$	8	9	11	10

Gray code order!
00,01,11,10

★ Key Exam Point: K-Map Wrap-Around

The K-Map is a **toroid** — edges wrap around! The left column is adjacent to the right column, and the top row is adjacent to the bottom row. Always check for wrap-around groups — they often give the biggest simplifications.

K-Map Group Size → Literal Reduction

Group Size	Literals Eliminated	Example (4 var)
1 cell	0	All 4 vars remain
2 cells	1	3 vars in product term
4 cells	2	2 vars in product term
8 cells	3	1 var in product term
16 cells	4	Output always 1 ($F = 1$)

§ Master Summary

1. Propositions: T/F statements. Tautology=always T; Contradiction=always F.

2. Operators: $\neg, \wedge, \vee, \oplus, \rightarrow, \leftrightarrow$. Precedence: NOT>AND>OR>$\rightarrow$>$\leftrightarrow$.

3. Equivalence: Same truth table. Key laws: De Morgan, Distributive, Implication ($p \rightarrow q \equiv \neg p \vee q$).

4. Predicates: $P(x)$ becomes a proposition when x is assigned or quantified.

5. Quantifiers: $\forall$ (all), $\exists$ (some). Negate by flipping quantifier and predicate.

6. Nested: Order matters! $\forall x\, \exists y \not\equiv \exists y\, \forall x$.

7. SAT: Satisfiable $\Rightarrow \exists$ assignment making formula true. SAT is NP-complete.

8. Inference: Modus Ponens, Modus Tollens, Hypothetical Syll., Disjunctive Syll., Resolution.

9. Normal Forms: CNF = AND of ORs. DNF = OR of ANDs. Every formula has both.

10. Circuits: NOT, AND, OR, NAND(universal), NOR(universal), XOR. Build from boolean expression.

11. K-Map: Group 1s in powers of 2; largest groups; Gray code order; wrap-around; eliminate literals.

EXAM PRACTICE — 20 QUESTIONS

MCQ (Q1–10) + Written (Q11–20) ★ Good luck! ★

PART A — Multiple Choice (Q1–10)

Q1. Which of the following is **NOT** a proposition?
(a) $2 + 2 = 5$ (b) "Close the door!" (c) "Paris is in France" **(d) b only**
Answer: (b) — commands and questions are not propositions.

Q2. $p \to q$ is logically equivalent to:
(a) $p \wedge \neg q$ **(b)** $\neg p \vee q$ (c) $\neg q \to p$ (d) $q \to p$

Q3. Which row makes $p \to q$ **FALSE**?
(a) $p = F, q = F$ (b) $p = F, q = T$ **(c)** $p = T, q = F$ (d) $p = T, q = T$

Q4. De Morgan's law states $\neg(p \vee q)$ is equivalent to:
(a) $\neg p \wedge \neg q$ (b) $\neg p \vee \neg q$ (c) $p \wedge q$ (d) $\neg p \to q$

Q5. The negation of $\forall x\, P(x)$ is:
(a) $\forall x\, \neg P(x)$ **(b)** $\exists x\, \neg P(x)$ (c) $\exists x\, P(x)$ (d) $\neg \forall x\, P(x)$ (unsimplified)

Q6. Which rule of inference is: "$p \vee q$; $\neg p$; $\therefore q$"?
(a) Modus Ponens (b) Addition **(c) Disjunctive Syllogism** (d) Resolution

Q7. A formula in **CNF** looks like:
(a) $(p \wedge q) \vee (r \wedge s)$ **(b)** $(p \vee q) \wedge (r \vee s)$ (c) $p \vee q \vee r$ (d) $p \wedge q \wedge r$

Q8. Which gate produces output 0 only when ALL inputs are 1?
(a) NOR **(b) NAND** (c) AND (d) XOR

Q9. In a K-Map, groups must have size equal to a **power of 2**. Which grouping is **invalid**?
(a) 1 cell (b) 4 cells **(c) 3 cells** (d) 8 cells

Q10. $\forall x\, \exists y\, (x + y = 0)$ over the integers means:
(a) There is one y that works for all x
(b) For every integer x, there is some integer y such that $x + y = 0$
(c) All integers sum to 0
(d) There exist x and y whose sum is 0

PART B — Written Questions (Q11–20)

Q11. [Truth Table]
Construct a truth table for $(p \to q) \wedge (\neg q \vee r)$. Identify all rows where the formula is true.

Q12. [Logical Equivalence]
Prove or disprove: $\neg(p \to q) \equiv p \wedge \neg q$.
(Hint: expand $p \to q$ using $\neg p \vee q$, then negate.)

Q13. [De Morgan's]
Simplify using De Morgan's and equivalence laws:
$\neg(\neg p \vee (q \wedge \neg r))$

Q14. [Predicate Logic — Translation]
Let $S(x)$ = "x is a student", $P(x)$ = "x passed the exam". Translate:
(a) Every student passed the exam.
(b) Some student did not pass.
(c) No student failed.
(d) At least one student passed.

Q15. [Nested Quantifiers — Negation]
Write the negation (pushed all the way in) of:
$\exists x \, \forall y \, (P(x, y) \to Q(y))$

Q16. [Rules of Inference]
Determine whether the following argument is valid. State which rules you use:
Premises: "If Alice studies, she passes." "Alice does not pass."
Conclusion: "Alice did not study."

Q17. [Normal Forms — DNF]
Convert to DNF: $F = (p \to q) \wedge r$

Q18. [Normal Forms — CNF]
Convert $F = p \vee (q \wedge r)$ to CNF. Show all steps.

Q19. [K-Map Minimisation — 3 Variable]
Minimise $F(A, B, C) = \sum m(1, 3, 5, 7)$ using a K-Map. Draw the K-Map, show groupings, write the minimised SOP expression.

Q20. [Circuit Design]
(a) Draw the logic circuit for $F = A \cdot B + \bar{C}$

(b) Identify which gates are used
(c) Create the truth table for F with inputs A, B, C

Answer Hints & Solutions

- **Q11:** Build 8-row table for 3 vars. Formula is T when both conjuncts are T.
- **Q12:** $\neg(p \to q) = \neg(\neg p \vee q) = p \wedge \neg q$ by De Morgan. TRUE, they are equivalent.
- **Q13:** $\neg(\neg p \vee (q \wedge \neg r)) = p \wedge \neg(q \wedge \neg r) = p \wedge (\neg q \vee r)$
- **Q14:** (a) $\forall x(S(x) \to P(x))$ (b) $\exists x(S(x) \wedge \neg P(x))$ (c) $\forall x(S(x) \to P(x))$ (d) $\exists x(S(x) \wedge P(x))$
- **Q15:** $\forall x\, \exists y\, \neg(P(x,y) \to Q(y)) = \forall x\, \exists y\, (P(x,y) \wedge \neg Q(y))$
- **Q16:** Valid. Let a =studies, p =passes. Premises: $a \to p$, $\neg p$. By **Modus Tollens**: $\therefore \neg a$.
- **Q17:** $p \to q \equiv \neg p \vee q$. So $F = (\neg p \vee q) \wedge r$. DNF: $(\neg p \wedge r) \vee (q \wedge r)$
- **Q18:** Distribute: $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$. Already CNF!
- **Q19:** All odd minterms $= B = 1$ column pairs in K-Map. Group all four: $F = C$ (the variable C selects all 1s here — check your specific encoding). Actually $m(1,3,5,7)$ all have $C = 1$, so $F = C$.
- **Q20:** Circuit: AND(A,B) feeds OR; NOT(C) feeds OR. Gates: AND, NOT, OR. Truth table: 8 rows, F=1 when (A=B=1) OR C=0.

All the best for your exam! ★

Logic · Predicates · Circuits · K-Maps — you have everything you need